

RIDESHIELD - ONE-PAGE TECHNICAL PROGRESS SHEET

IoTrix 2.0 Semi-Final | Track A - Embedded IoT System | 5 October 2026

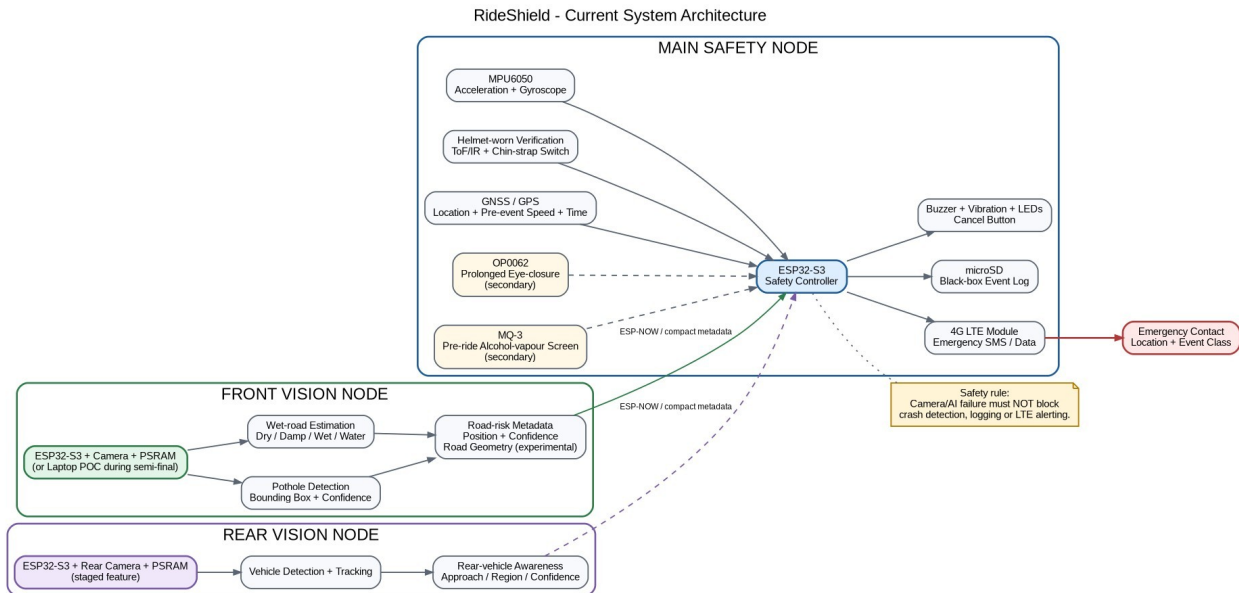
PROBLEM

Motorcycle riders can face delayed emergency response after a crash, road hazards such as potholes and wet surfaces, and limited rear awareness. The core safety functions must also remain operational even if non-critical camera/AI modules fail.

PROPOSED SOLUTION

RideShield uses an independent ESP32-S3 safety node for helmet-worn verification, MPU6050 crash sensing, rider cancellation, GPS/LTE emergency handling, microSD black-box logging and warnings, with separate front/rear vision nodes for road and rear-vehicle awareness.

ARCHITECTURE



CURRENT PROGRESS

- Modular architecture finalized: independent ESP32-S3 safety core with separate front/rear vision nodes.
- Core safety firmware implemented in Wokwi with NOT WORN -> READY -> VERIFYING -> COUNTDOWN -> EMERGENCY state flow.
- Helmet-worn gating implemented using combined head-presence + strap state; invalid helmet state rejects crash processing.
- MPU6050 crash logic implemented: candidate trigger, 1.2 s multi-factor verification, peak acceleration/gyro, pre-event speed and post-event stillness.
- 10 s non-blocking rider-cancel workflow and prototype 0-100 Crash Event Severity Index implemented.
- GPS current-fix/last-known/no-fix logic, LTE offline queue + automatic retry, and microSD event/black-box logging implemented.
- Automated Wokwi end-to-end scenario completed successfully; pothole CV workspace/model and demo inference pipeline remain separate.

TECHNOLOGY STACK

- Implemented PoC: ESP32-S3, MPU6050, microSD, LEDs/buzzer/buttons, Arduino C+ +, PlatformIO, Wokwi Automation, I2C and SPI.
- Interface-level simulation: head/strap status, GPS fix + speed, and LTE/network availability.
- Vision development: Python, OpenCV and lightweight object-detection/classification models; compact inter-node metadata transfer planned.

TESTING / VALIDATION

- Helmet validation: head detected + strap open -> crash request rejected; valid helmet state -> READY.
- Crash/cancel test: ~6.08 g, ~700°/s, 60 km/h, 4 verification signals -> rider cancel accepted -> 48/100 Moderate event logged.
- Emergency test: ~10.05 g, ~1000°/s, 60 km/h, 4 signals -> no response -> 73/100 High-Energy event -> current GPS fix -> simulated SMS success.
- Failure test: GPS loss -> last-known location; LTE loss -> alert queued; LTE restored -> automatic retry succeeded.
- Storage test: events.csv recorded cancelled/confirmed summaries; blackbox.csv preserved pre-event acceleration, gyro, speed and helmet state.

LIMITATIONS / RISKS

- Current validation is simulation-based; real sensor noise, GNSS RF, LTE RF/power, battery behaviour and helmet mechanics are not yet validated.
- MPU6050 saturates above ±16 g; thresholds, normalization limits and severity weights are prototype values requiring calibration.
- Crash Event Severity Index describes event characteristics, not medical injury severity; camera wetness estimation does not measure tyre-road friction.
- Helmet-mounted camera motion limits fixed geometry/rear-zone assumptions; local Sri Lankan data are required for robust vision performance.

NEXT STEP BEFORE FINAL

- Calibrate crash thresholds with controlled labelled tests and integrate physical MPU6050, GNSS, LTE, helmet-worn sensing and stable power delivery.
- Quantify pothole/wet-road model accuracy using local road data, optimize edge deployment, then integrate rear-awareness and secondary rider-monitoring features.